

Homework Section 14.8

1. Use Lagrange Multipliers to find the maximum and minimum values of the function subject to the given constraint(s).
 - a) $f(x, y) = x^2 - y^2$; $x^2 + y^2 = 1$
 - b) $f(x, y) = x^2 y$; $x^2 + 2y^2 = 6$
 - c) $f(x, y, z) = 2x + 6y + 10z$; $x^2 + y^2 + z^2 = 35$
 - d) $f(x, y, z) = xyz$; $x^2 + 2y^2 + 3z^2 = 6$
 - e) $f(x, y, z) = x^4 + y^4 + z^4$; $x^2 + y^2 + z^2 = 1$
 - f) $f(x, y, z) = x + 2y$; $x + y + z = 1$, $y^2 + z^2 = 4$
2. Find the extreme values of $f(x, y) = e^{-xy}$ on the region $R = \{(x, y) \mid x^2 + 4y^2 \leq 1\}$.
3. Use Lagrange Multipliers to find the points on the surface $z^2 = xy + 1$ that are closest to the origin.
4. A cardboard box without a lid is to have a volume of 32000 cubic centimeters. Use Lagrange Multipliers to find the dimensions that minimize the amount of cardboard used.